

Aligned Mathematical Structures

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$<\chi,\chi'>=\prod_{r=1}^N\delta_{n_r n'_r}\cdot\prod_{r=1}^M\delta_{m_r m'_r}\cdot\prod_{r=1}^Q\delta_{q_r q'_r}\cdot\prod_{r=1}^T\delta_{t_r t'_r}\cdot\prod_{r=1}^4\delta_{\nu_r \nu'_r}.$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned}\mathcal{Q}_T^k(x_{1:T-1}^k) &\stackrel{Lemma\ ??}{=} \theta_T^k \stackrel{(\ ??)}{=} \rho_T(\mathfrak{Q}_T^{k-1}(x_{1:T-1}^k, \xi_T)) \\ &= \rho_T(\mathfrak{Q}_T(x_{1:T-1}^k, \xi_T)) \stackrel{(\ ??)}{=} \mathcal{Q}_T(x_{1:T-1}^k).\end{aligned}$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned}(\partial_1 b)(p_1,p_2) &:= p_1 + \frac{p_1 p_2}{1-p_2} \Phi_s + \frac{(p_1^2 + p_2^2) p_1 p_2}{1-p_2} \frac{\Phi'_s}{s^2} \\ (\partial_2 b)(p_1,p_2) &:= p_2 + \frac{p_2^2}{1-p_2} \Phi_s + \frac{1}{2} \frac{p_1^2 + p_2^2}{(1-p_2)^2} \Phi_s + \frac{(p_1^2 + p_2^2) p_2^2}{1-p_2} \frac{\Phi'_s}{s^2}.\end{aligned}$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$(VI)' \quad \Psi_{\mathbf{D}} = \begin{pmatrix} a_1 + \mathrm{j}b_1 \\ a_2 + \mathrm{j}b_2 \\ a_3 + \mathrm{j}b_3 \\ a_4 + \mathrm{j}b_4 \end{pmatrix} \quad \leftrightarrow \quad \Psi_{\mathbf{C}}^{SR} = \begin{pmatrix} -b_4 + \mathrm{i}b_2 \\ b_3 - \mathrm{i}b_1 \\ a_4 - \mathrm{i}a_2 \\ -a_3 + \mathrm{i}a_1 \end{pmatrix}$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned}\frac{\delta}{\delta h_{i'j'}(x')}G_{a'b'cd}(y) &\equiv g_{a'b'cd}^{i'j'}(y) \, \delta(y,x') \, , \\ \frac{\delta}{\delta h_{kl}(x)}\frac{\delta}{\delta h_{i'j'}(x')}G_{a'b'cd}(y) &\equiv g_{a'b'cd}^{i'j'kl}(y) \, \delta(y,x') \, \delta(y,x) \, .\end{aligned}$$